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Report on the Examination

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REPORT ON THE EXAMINATION: INTERNATIONAL – JANUARY 2026

This paper was similar to those of previous years in terms of specification coverage and demand. Students had many opportunities to demonstrate their knowledge, skills and understanding across a range of topics. The usual variety of question styles including short answers, single and multi-step calculations, extended writing and multiple choice were used. The questions were set in a range of both familiar and unfamiliar contexts.

The balance of assessment objectives and the mathematical demand was also similar to those of previous series.

There was no evidence of any student running out of time.

QUESTION A1.1

This question proved to be very straightforward, with over 80% of students receiving full marks. There were several different routes seen, all of which gained credit. The mark scheme was designed so that incorrect answers could get partial credit if the working out was sufficiently clear. One common error was to use the volume of the nitrogen gas rather than the volume of sweets.

QUESTION A1.2

This question was also answered very well, with over 90% of students achieving full marks. The most common first step was to determine the number of moles. Many of those who chose to work out the number of particles did not make correct use of Avogadro's number in their final answer.

QUESTION A1.3

This calculation was also completed successfully by over 90% of students and it discriminated reasonably well. There were several alternative methods seen with the use of the idea that the pressure divided by absolute temperature is constant being the most common. Some students did not use the correct conversion to kelvin, but these were still able to obtain the alternative mark for MP1.

QUESTION A1.4

The explanation required in this question proved to be much more challenging with only about 5% of students gaining all three marks. The question required students to refer to the movement of particles but this was rarely done with much success. Students should be encouraged to refer to the **mean** speed or momentum of particles in a gas. Most commonly students referred to pressure with little reference to the different forces on the lid. This was only given a compensatory mark if nothing else of credit was seen. This mean that over 80% of students could pick up at least one mark.

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QUESTION A2.1

This question was answered correctly by more than 80% of students. No reasoning was expected in the answer so any that was provided was ignored by examiners.

QUESTION A2.2

Most students were able to complete the multistep calculation required here. The calculator value was provided in the mark scheme so that early rounding was penalised. It was clear from the few incorrect answers that were seen that some students struggled with having two different materials being heated.

QUESTION A2.3

This question was much more challenging, with fewer than 60% of students gaining full marks. Many students simply related the 36kJ to the mass multiplied by the latent heat without any reference to the specific heat capacity of the water.

QUESTION A2.4

Only about 10% of students were awarded both marks for this question, with about half getting at least one mark. The most common error was stating a suitable change without giving a suitable reason. The question clearly states 'state and explain'. Students should be encouraged to use the command words to help structure their answer and ensure that they have answered questions fully.

QUESTION A3.1

This five mark questions produced a very good spread of marks, with about 20% of students being given full marks. There were several methods, all of which gained credit. The most common answer was to plot R against $A^{1/3}$. This had the advantage that the units were easily added to the axis label. Those choosing a log plot often failed to get the label mark as they did not include the unit within the argument of the log i.e. log(R/fm). Many students struggled with the scales of the graph. Examiners expected to see scales that were easy to read and used at least half of the graph paper. Scales going up in steps of 3 were not rewarded, for example. The final mark also proved to be difficult for many students. The analysis required depended on the graph chosen. For example examiners were expecting to see a reference to $y = m x + c$ with a demonstration of $c = 0$ for the R against $A^{1/3}$ graph. Many students simply showed that there was a constant relationship using the values in the table. This was not rewarded as the question was about the graph, not the data.

QUESTION A3.2

About 75% of students gained both marks for this calculation. Examiners expected to see evidence of the use of the graph. Most commonly this was done by a read-off. Again those who chose a log-log plot tended to have more difficulty scoring in this question.

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QUESTION A3.3

The fact that nearly 70% of students were given full marks is an indication that students were well prepared for this question. Common errors included using the incorrect charge for the alpha particle and squaring the distance between them. The mark scheme was designed so that clearly set out work could be given some credit even if the final answer was wrong.

QUESTION A3.4

Surprisingly, only about 20% of students obtained the mark despite similar questions being asked in the past. One common error was due to some confusion with Rutherford scattering. Over 10% of students made no attempt at this question.

QUESTION A4.1

Over 70% of students were able to give a suitable definition of binding energy. A common error was to simply mention constituent particles rather than nucleons, or protons and neutrons.

QUESTION A4.2

The majority of students were able to draw the general shape of the curve. Many fewer were able to show the correct values for the peak or for the end of the line, despite there being quite a large tolerance.

QUESTION A4.3

About 25% of students obtained both marks for this question. Many students had difficulty expressing their answers so that it was not clear whether they were referring to binding energy or binding energy per nucleon. At least one mark was awarded to over 70% of students. Those who only gained one mark were often credited for the first marking point.

QUESTION A4.4

Many students were not awarded the mark for this question because their answers were either too vague or incomplete. Simply stating that the nucleus contains a proton was insufficient. Examiners were looking for the idea that there were no other nucleons in the nucleus.

QUESTION A4.5

Almost 70% of students were able to complete this calculation correctly. The mark scheme was designed to reward intermediate marks to students who did not get the correct answer. This also meant that it was one of the best discriminating questions on the paper. The most common error was to miss out the conversion of binding energy per nucleon to binding energy.

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QUESTION A5.1

This straightforward calculation was one of the most accessible on the paper. The most common error was to divide their angular speed by two pi.

QUESTION A5.2

When answering this question, the most successful students set out their answers clearly so that each step could be seen and rewarded. Over 60% of students were given full marks. The most common errors were to mix up forces and torques and to use the frictional torque incorrectly. This question discriminated very well.

QUESTION A5.3

Over 80% of students successfully applied the conservation of angular momentum to answer this question and gain full marks. Those who only gained one mark often used the wrong moment of inertia in their calculation.

QUESTION A5.4

This question also discriminated very well. Several approaches were seen and all were given credit. The mark scheme was generic so that two marks were available for the approach the students chose to take. The final mark was given for the two correct answers. Several students were able to determine the time correctly but made an error when calculating the angular displacement. These students could still get two out of the three marks provided their working was clear.

QUESTION A6.1

Surprisingly few students gained the mark for this question. This was largely due to writing the current to only one significant figure. Most students gave 0.04 (A) but examiners needed to see 0.040 (A). There was a small acceptable range.

QUESTION A6.2

This question was very accessible with almost 90% of students gaining both marks. There were power of ten errors that limited the mark for some students to one.

QUESTION A6.3

This question discriminated reasonably well and produced a good spread of marks. The most common error was to use 75% rather than 25% in the calculation. The mark scheme was designed to reward each step in the calculation so that nearly 90% of students were able to get at least two marks.

QUESTION A7

This question produced a good spread of marks and discriminated reasonably well. Very few full mark answers were seen, however. This was often due to a lack of detail in the third area, a discussion of the factors that determine the maximum power. The most successful students used the maximum power equation and defined each of the terms.

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Students who merely suggested that the difference in height of the reservoirs was the factor did not score in this area. There was also evidence of some confusion of a pumped storage system and a simpler hydroelectric scheme. Students should be encouraged to read the question carefully and use the bullet points provided to make sure they answer the question fully.

SECTION B

None of the multiple choice questions were particularly demanding.

B16 was the hardest question but was still answered correctly by over 75% of students. The most popular incorrect answer to this question was A. It may be that these students assumed that the angular speed rather than the angular momentum of each disc was the same.

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